



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech. (Computer Science & Engg.) (CBCS) Degree Examinations

May - June 2024

23CSE222 – GENERAL PURPOSE COMPUTATION ON GPU

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the architecture of CUDA-Enabled NVIDIA GeForce 8800 GPUs with neat block diagram.	10	L*2	1	1
	b) Describe the purpose of following built-in variables in CUDA: i) gridDim ii) blockDim iii) blockIdx iv) threadIdx	6	L1	2	1
	c) Differentiate between the features of CUDA and OpenCL.	4	L2	2	1
2.	a) Explain the fixed function NVIDIA graphics pipeline with neat block diagram	10	L2	1	1
	b) What are the different types of parallelism? Briefly explain each type.	6	L2	1	1
	c) A grid is having 5 blocks with 64 threads per block that are organized as 8×8 array. Write the kernel execution configuration for this 1D grid having 2D blocks.	4	L1	2	2
3.	a) Differentiate between CPU and GPU architectures. With block diagram explain the generalized structure of CUDA-enabled GPU.	10	L2	1	1
	b) Write the expression for the Global Thread Index for a CUDA kernel having 2D Grid of 2D Thread Blocks.	6	L3	2	1
	c) With an example briefly explain the features of heterogeneous architecture.	4	L2	1	1
Unit – II					
4.	a) Outline the steps involved in executing an OpenCL kernel. Name the APIs used in each step.	10	L2	3	1
	b) Write a CUDA program to add two vectors using 1-dimensional grid of multiple 1D Blocks, where each block consists of multiple threads.	6	L3	3	1
	c) Explain the purpose of __syncthreads () in CUDA.	4	L2	4	1
5.	a) Explain the OpenCL memory hierarchy with a neat block diagram.	10	L2	4	1
	b) Write a CUDA program to multiply two matrices using a grid of single 2D Block.	6	L3	3	1
	c) Explain the Kernel, Grid, Block, and Thread with respect to CUDA.	4	L1	3	1
6.	a) Write a CUDA program for matrix multiplication using shared memory.	10	L3	4	1
	b) Explain the cudaMalloc() and cudaMemcpy() functions with suitable examples.	6	L2	4	1
	c) Briefly explain the transparent scalability feature of CUDA.	4	L1	3	1
Unit – III					
7.	a) List and explain the features of various debugging tools used in CUDA.	10	L2	5	1
	b) Outline the purpose of profiling a GPU application.	10	L2	5	1
8.	a) Explain the factors involved in selecting number of blocks per grid and the block size.	10	L2	5	1
	b) Explain how control divergence on GPU application affects the performance. Suggest methods to minimize the number of divergent warps.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
